EMMA — Integration & Mobile

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Next.js dashboard · API connections · Expo app maintenance · GitHub
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API Contract v1.0
Integration & Mobile role
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> **Your role in one sentence:** Build the Next.js DRRMO operator dashboard, wire it to Alliyana's Python service and Ryu's Laravel endpoints, maintain the existing Expo citizen app, and keep the GitHub clean enough for judges to clone and run in 10 minutes.

WHAT YOU ARE BUILDING — NEXT.JS DRRMO OPERATOR DASHBOARD

New interface — distinct from citizen app

What the dashboard shows

- Multilingual situation report input field
- Language selector (Filipino / Bahasa / Vietnamese / Thai / English)
- Agent pipeline status (which agents are processing)
- Agent outputs with reasoning trace and confidence score per agent
- Human Gate — Approve / Modify / Reject buttons
- Decision confirmation with audit trail note

Tech stack

What to use

Framework: Next.js (App Router or Pages — your call)

Styling: TailwindCSS

Language: TypeScript

Hosting: Azure SEA (free student credits)

Environment variables (.env.local)

```
LARAVEL_BASE_URL=http://localhost:8000
PYTHON_SERVICE_URL=http://localhost:8001
```

OPERATOR FLOW — WHAT THE DASHBOARD CALLS AND WHEN

1 On page load — health check

Call **GET /health** on Alliyana's Python service. Show green status indicator if agents are running. Show a warning banner if the service is down.

```
GET {PYTHON_SERVICE_URL}/health
→ { "status": "ok", "agents": 5, "llm": "claude-haiku-4-5" }
```

2 Operator submits a situation report

First, POST to Ryu's Laravel to register the report and get a report_id.

```
POST {LARAVEL_BASE_URL}/api/situation-reports
Body: { "report_text": "string", "lgu_id": "string", "submitted_by": "string", "timestamp": "ISO datetime" }
→ { "report_id": "string", "status": "received" }
```

3 Send to Python agent service

POST to Alliyana's service with the report_id and text. Display agent outputs progressively as they arrive. Show each agent card when its output is received.

```
POST {PYTHON_SERVICE_URL}/process
Body: { "report_id": "string", "report_text": "string", "lgu_id": "string", "timestamp": "ISO datetime" }
→ full agent output object (see schema below)
```

4 Show Human Gate after Handoff Coordinator completes

Display the Handoff Coordinator's merged recommendation with reasoning. Show the three decision buttons. Do NOT dispatch any action until the operator explicitly clicks one.

5 Log the operator decision to Laravel

After operator clicks Approve / Modify / Reject, POST to Ryu's Laravel with the full agent output + decision. This completes the audit trail.

```
POST {LARAVEL_BASE_URL}/api/agent-logs
Body: {
  "report_id": "string",
  "agent_outputs": { ... full agent output },
  "operator_decision": "approved | modified | rejected",
  "operator_id": "string",
  "decided_at": "ISO datetime"
}
```

AGENT OUTPUT SCHEMA — WHAT ALLIYANA RETURNS, WHAT YOU DISPLAY

```
{
  "report_id": "string",
  "intake":        { "language_detected": "", "location": "", "hazard_type": "", "population_affected": 0,
"urgency": "", "key_flags": [], "confidence": 0 },
  "vulnerability": { "tier1": [], "tier2": [], "recommendation": "", "confidence": 0 },
  "resource":      { "gaps": [], "available": [], "recommendation": "", "confidence": 0 },
  "routing":       { "primary_status": "", "alternative": "", "eta_minutes": 0, "risk": "",
"recommendation": "", "confidence": 0 },
  "pattern":       { "prior_event": "", "insight": "", "cross_border": "", "confidence": 0 },
  "handoff":       { "action": "", "reasoning": "", "confidence": 0 }
}
```

Show each agent as a card with its confidence score badge. The handoff section triggers the Human Gate display.

EXISTING EXPO APP — YOUR MAINTENANCE RESPONSIBILITIES

Keep working

All 30+ existing screens

Do not change any existing functionality. The citizen app is a complete product that demonstrates the field side of EMMA. It runs alongside the operator dashboard — both are shown in the LGU pitch.

Connect new endpoints

Link to Ryu's new APIs

If any new Laravel endpoints Ryu adds are relevant to existing citizen app screens, connect them. Specifically: the vulnerability flags might surface in family/member screens.

GitHub hygiene

10-minute clone test

README must let a judge clone the repo and run the full demo in 10 minutes. One command ideally. Test it on a clean machine before submission. This is a Gate 2 scoring item.

GITHUB STRUCTURE — WHAT JUDGES WILL CLONE

emmafrontend

Existing Expo citizen app

Stays as-is with updated README. Add setup instructions for running with the new backend endpoints.

emmabackend

Laravel + Python agent service

```
emmabackend/
├── laravel/    ← Ryu's existing + new
endpoints
├── agents/     ← Alliyana's Python service
└── README.md   ← covers both, single setup
guide
```